COMPUTER FORENSICS

Disk Drives and Solid-State Storage Devices

Examining HDD and SSD architecture through a digital-evidence lens

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OVERVIEW

Learning Objectives

By the end of this session, you'll understand how computers store data physically — and why that matters when investigating digital evidence.



Understand storage devices used in computers 01



Learn the architecture of HDD and SSD 02



Compare HDD and SSD 03



Understand forensic challenges in SSDs 04



Explore the importance of storage devices in digital investigations 05

FOUNDATIONS

Storage Devices in Computer Forensics

Think of it as a digital filing cabinet — unlike RAM, it keeps data even after the power is switched off.



What is a Storage Device?

Hardware used to store digital information permanently — so files, photos, and logs survive a restart or shutdown.

Importance in Computer Forensics

- Primary source of digital evidence
- Stores user and system data
- Used to recover deleted information

Evidence Examples



Documents



Images & Videos



Emails



Browser History



Chat Messages



System Logs

MAGNETIC STORAGE

Hard Disk Drive (HDD)

Analogy: much like a record player — spinning platters plus a moving arm that reads and writes magnetically.

Definition

Stores data magnetically on rotating platters.

Main Components

- **Platters** – spinning magnetic disks that hold the data
- **Read/Write Head** – reads and writes data as platters spin beneath it
- **Spindle Motor** – spins the platters at high, constant speed
- **Actuator Arm** – moves the head to the right position on the platter
- **Controller Board** – manages data flow between the drive and the computer

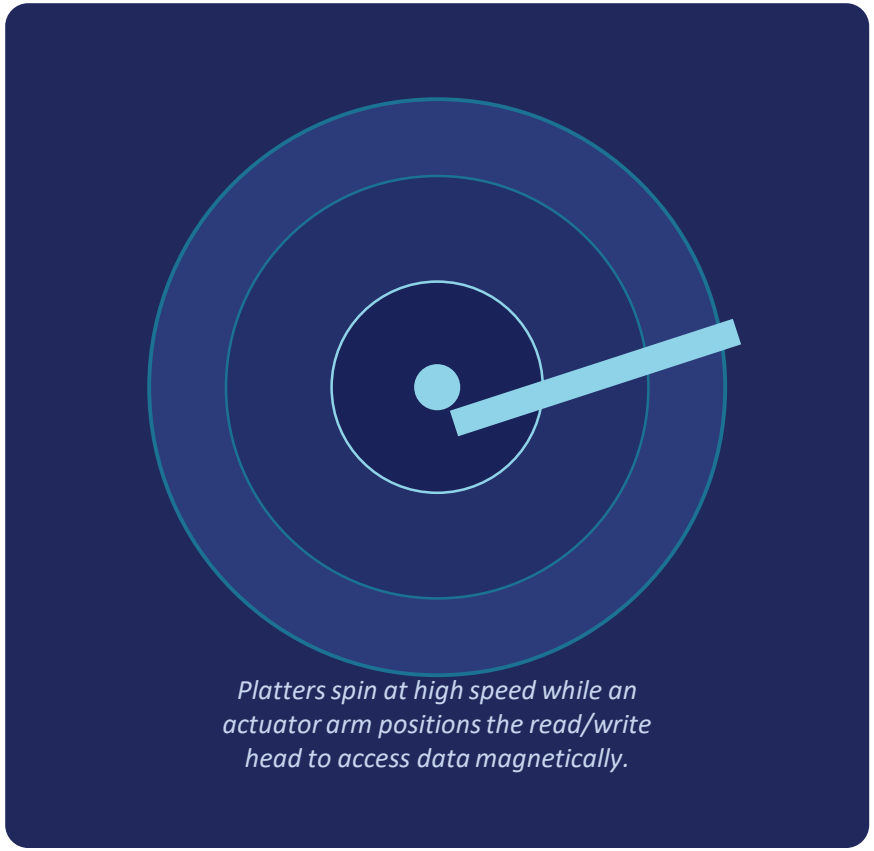
Characteristics

Mechanical device

Large storage capacity

Lower cost

Slower performance



Platters spin at high speed while an actuator arm positions the read/write head to access data magnetically.

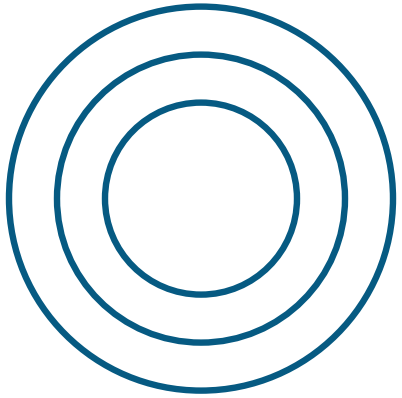
MAGNETIC STORAGE

How HDD Stores Data

A track is a lane on the platter, a sector is one parking spot on that lane, and a cluster is a group of spots reserved for one file.

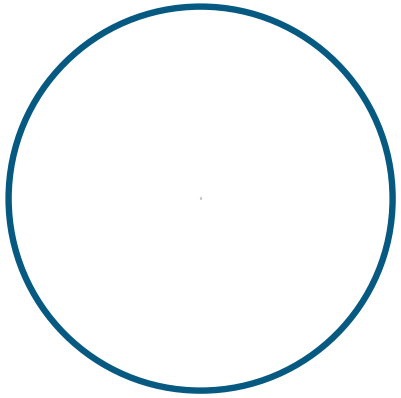
HDD Data Organization

Data is organized into:



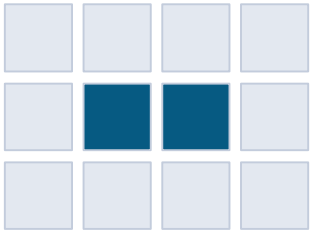
Tracks

Concentric circular paths on the platter



Sectors

Tracks divided into fixed-size segments



Clusters

Groups of sectors allocated to files

Forensic Importance: Deleted files often remain on the disk until overwritten, making recovery possible.

ELECTRONIC STORAGE

Solid-State Drive (SSD)

Analogy: similar to a USB flash drive — electronic memory chips, no motors or moving parts.

Definition

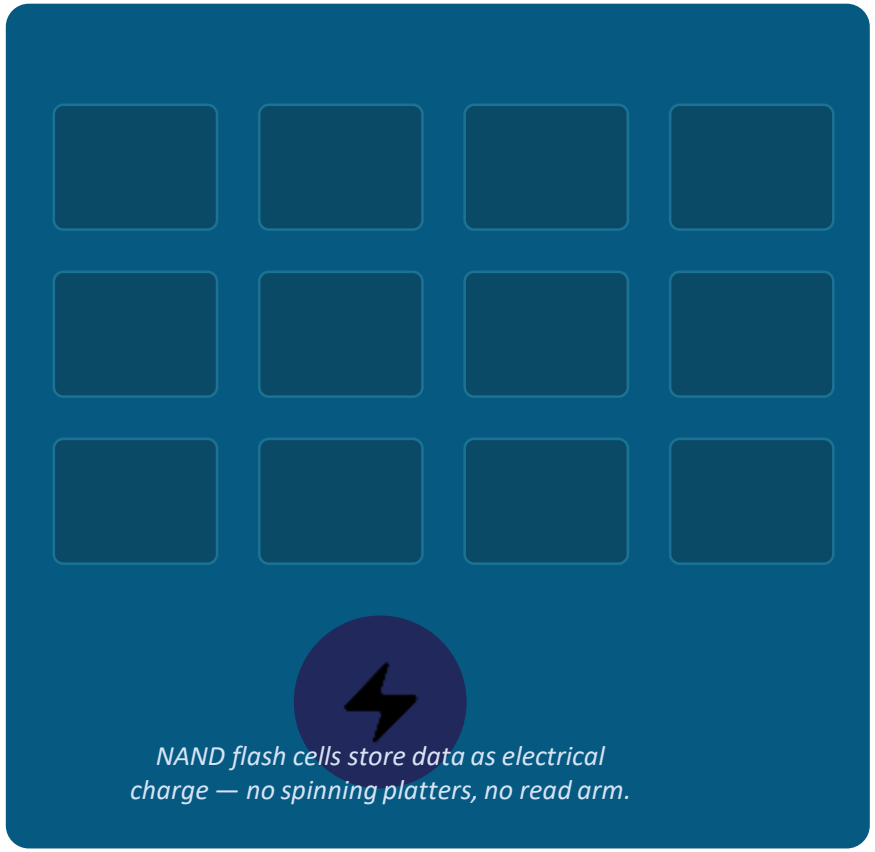
Stores data electronically using NAND Flash Memory.

Components

- **NAND Flash Memory –** chips that hold data as stored electrical charge
- **SSD Controller –** the “brain” that manages where data is written and read
- **Cache Memory –** small, fast buffer that speeds up frequent operations

Characteristics

- **No moving parts**
- **Faster performance**
- **Lower power use**
- **Shock resistant**



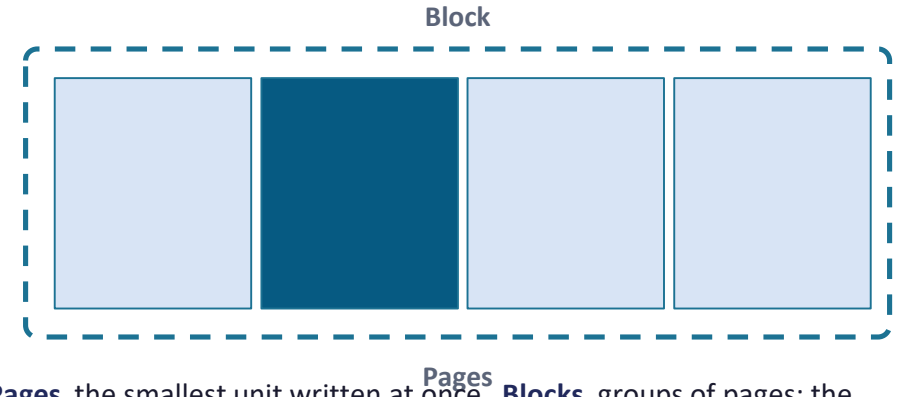
NAND flash cells store data as electrical charge — no spinning platters, no read arm.

ELECTRONIC STORAGE

How SSD Stores Data

Pages are like sheets in a notebook you can fill one at a time — but you can only erase a whole block (a full section) at once.

Data Organization



Pages the smallest unit written at once. **Blocks** groups of pages; the smallest unit erased at once.

SSD Controller Functions

Address Translation

Maps logical addresses to physical NAND locations

Wear Leveling

Spreads writes evenly to extend memory lifespan

Garbage Collection

Reclaims space from invalid, stale data pages

Error Correction

Detects and fixes bit-level read/write errors

COMPARISON

HDD vs SSD

Same job — storing your data — but very different technology, speed, and price.

	HDD	SSD
Storage Method	Magnetic Storage	NAND Flash Memory
Moving Parts	Moving Parts	No Moving Parts
Speed	Slower	Faster
Cost	Lower Cost	Higher Cost
Data Recovery	Easier Data Recovery	Difficult Recovery

FORENSIC CHALLENGES

Why SSDs Are Challenging

These background processes make SSDs fast and long-lasting — but they actively work against forensic data recovery.



TRIM Command

- Marks deleted blocks as unused
- Reduces chances of recovering deleted files



Garbage Collection

- Cleans unused memory blocks
- Permanently removes deleted data



Wear Leveling

- Distributes writes across memory
- Changes physical data locations

COMPARISON

HDD vs SSD — Forensic Perspective

Putting it together: the very features that make SSDs efficient are what make deleted evidence so hard to recover.

Feature	HDD	SSD
Deleted File Recovery	Easier	Difficult
TRIM	No	Yes
Garbage Collection	No	Yes
Wear Leveling	No	Yes
Forensic Complexity	Low	High

SSD features designed for performance and longevity directly complicate digital forensic recovery.

SUMMARY

Key Takeaways



HDD stores data magnetically on spinning platters — slower, cheaper, and easier to recover deleted files from.



SSD stores data electronically in NAND flash — faster and more durable, but built-in maintenance erases data permanently.



TRIM, garbage collection, and wear leveling are the core reasons SSD forensic recovery is far harder than HDD.



Knowing how a drive stores and deletes data helps investigators judge what evidence can realistically be recovered.